

Fall 2025.

**Physics 501. Methods of Theoretical Physics I.
Yurii Shylnov.**

Homework # 4.

Due to October 21st, 2025.

Read the corresponding chapters. If you forgot anything necessary from your core math courses, find the information in the textbook, online or whatever works for you. Solve the problems. It should be done in a very detailed and logical way. You need to understand every step of your solution.

1. Bessel functions.

Arfken&Weber, Chapter 14. Problems 14.1.1, 14.1.2, 14.1.4, 14.1.6, 14.1.7, 14.1.9, 14.1.10, 14.1.19.

2. Modified Bessel functions.

Arfken&Weber, Chapter 14. Problems 14.5.1, 14.5.2, 14.5.6, 14.5.7, 14.5.14, 14.5.15.